

AYR, Australia

WMO: 952950

Lat: 19.6169S

Lon: 147.3758E

Elev: 56

StdP: 14.67

Time Zone: 10.00 (E10)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	47.5	50.4	32.3	26.8	65.8	37.3	32.7	63.7	26.3	77.4	18.5	71.6	4.4	200	0.340

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.3	91.5	76.5	89.9	76.2	88.7	75.8	81.5	86.0	79.9	85.6	78.9	84.6	9.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	157.7	83.2	78.3	147.4	82.5	77.3	142.3	81.8	45.2	86.6	43.4	85.7	42.4	84.8	92.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.0	14.3	41.9	96.4	2.5	2.7	40.1	98.4	38.6	100.0	37.2	101.5	35.4	103.5		
(5)			39.1	82.4	2.2	2.8	37.6	84.4	36.3	86.1	35.1	87.7	33.5	89.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.3	80.8	80.9	79.2	75.6	71.0	66.8	65.7	67.0	71.2	75.2	78.1	80.6
	DBStd	6.33	2.26	2.41	2.12	2.31	3.57	4.28	4.00	3.47	2.96	2.92	2.46	2.44
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	98	0	0	0	0	5	31	41	20	1	0	0	0
	CDD50	8876	955	865	905	768	652	505	487	526	637	782	844	950
	CDD65	3500	490	445	440	318	193	86	63	81	188	317	394	485
	CDH74	29686	4959	4427	3931	2221	1028	381	286	518	1314	2415	3359	4847
CDH80	9014	1768	1600	1256	505	111	15	5	30	208	626	1053	1837	
Wind	WSAvg	6.8	7.0	6.8	6.9	6.9	6.3	6.2	6.2	6.6	7.4	7.5	7.2	
Precipitation	PrecAvg	39.7	8.2	9.3	6.3	2.0	1.7	0.9	0.5	0.7	0.4	1.1	1.8	5.5
	PrecMax	69.1	37.6	25.6	21.4	16.9	7.5	8.3	2.2	4.5	2.5	9.3	8.5	19.9
	PrecMin	14.4	0.6	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4
	PrecStd	15.4	7.9	6.7	5.4	3.3	2.0	1.4	0.6	1.0	0.7	1.7	2.1	5.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	95.0	90.8	87.6	84.6	82.0	80.6	82.8	87.2	90.5	91.2	92.9
		MCWB	79.0	76.1	76.4	72.7	71.0	69.3	66.0	65.3	66.2	69.5	73.8	76.5
	2%	DB	90.3	91.4	88.9	85.5	82.7	79.6	78.6	80.4	84.1	86.7	88.2	90.7
		MCWB	77.5	77.2	76.2	72.9	70.2	67.8	65.0	65.5	68.3	72.3	73.7	76.3
	5%	DB	88.9	89.4	87.4	84.4	81.1	78.1	77.1	78.8	82.3	85.0	86.8	89.2
		MCWB	76.8	76.7	75.5	72.1	69.3	66.7	64.3	64.6	68.2	71.3	73.1	75.8
	10%	DB	87.5	87.8	86.0	83.2	79.6	76.4	75.5	77.1	80.6	83.5	85.4	87.9
		MCWB	76.2	76.3	75.0	71.8	68.4	65.2	63.6	63.7	67.2	70.2	72.5	75.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	85.8	82.1	83.4	82.3	76.7	76.2	71.1	71.7	74.2	76.2	79.2	85.6
		MCDB	86.0	86.7	84.9	82.7	81.3	77.9	75.2	78.1	80.7	83.8	84.6	85.9
	2%	WB	80.5	80.1	79.7	77.7	73.8	71.0	68.7	69.7	72.4	74.8	77.2	80.7
		MCDB	87.1	86.3	84.6	79.8	79.4	76.1	74.9	76.0	79.0	82.1	83.5	86.6
	5%	WB	79.2	79.2	78.1	75.3	71.7	69.0	67.2	67.8	70.8	73.7	75.9	78.9
		MCDB	85.3	85.1	83.8	80.2	78.0	75.1	74.0	75.0	78.3	81.1	82.6	85.1
	10%	WB	78.2	78.4	76.8	73.7	70.1	67.4	65.8	66.2	69.3	72.4	74.6	77.4
		MCDB	84.0	84.1	82.8	79.8	76.8	73.7	72.8	73.8	77.7	80.2	81.6	83.8
Mean Daily Temperature Range	5% DB	MDBR	13.3	13.3	14.2	15.9	18.1	19.5	21.0	21.7	20.1	17.3	15.0	14.5
		MCDBR	15.3	16.5	16.2	17.3	19.0	19.5	21.0	22.6	21.4	18.3	16.3	16.0
		MCWBR	5.5	6.0	6.2	7.3	8.8	9.3	9.6	9.7	8.8	6.3	5.1	5.2
	5% WB	MCDBR	12.7	12.6	13.1	13.2	16.6	16.6	16.3	18.9	18.1	15.9	13.3	13.5
		MCWBR	5.8	5.6	6.5	7.7	9.7	9.2	8.6	9.1	8.3	6.0	5.1	6.9
Clear-Sky Solar Irradiance	taub	0.403	0.396	0.388	0.364	0.345	0.332	0.321	0.334	0.362	0.389	0.395	0.403	
	taud	2.513	2.539	2.541	2.595	2.605	2.629	2.643	2.584	2.501	2.450	2.472	2.478	
	Ebn at Noon	299	298	292	287	282	280	287	293	295	296	299	299	
	Edh at Noon	36	35	34	30	27	26	26	30	34	38	37	37	
All-Sky Solar Radiation	RadAvg	2034	1902	1766	1591	1364	1231	1381	1642	1959	2147	2193	2186	
	RadStd	167	200	146	137	93	107	90	99	109	120	232	197	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page